MIL-T-18134C
29 May 1986
SUPERSEDING
MIL-T-18134B(NAVY)
14 August 1980

MILITARY SPECIFICATION

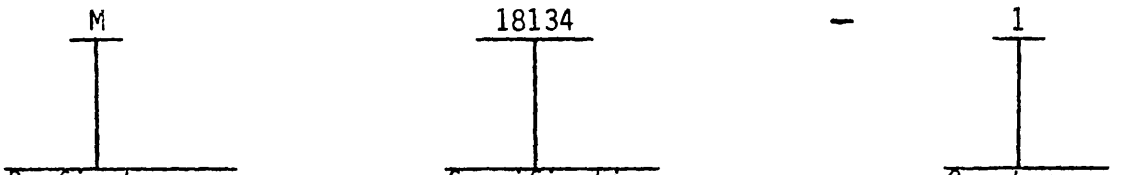
TORCHES, WELDING, HAND, OXYACETYLENE, LIGHT DUTY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of light duty oxyacetylene fueled hand welding torch.

1.2 Part number. The standard part number for the torch is:



2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, Memphis, Tennessee 38114-5297, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATION

MILITARY

MIL-W-45562 - Welding and Soldering Equipment, Supplies and Accessories, Packaging of

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

2.1.2 Other Government publications. The following other Government publication forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on the date of the solicitation.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Copies of specifications, standards, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment document which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/UL 123 - Oxy-Fuel Gas Torches.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B 124 - Copper and Copper Alloy Forging Rod, Bar, and Shapes.

ASTM B 133 - Copper Rod, Bar, and Shapes.

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ASTM D 572 - Test Method for Rubber Deterioration by Heat and Oxygen.

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

COMPRESSED GAS ASSOCIATION (CGA)

CGA E-1 - Standard Connections for Regulators, Torches, and Fitted Hose for Welding and Cutting Equipment.

(Application for copies should be addressed to the Compressed Gas Association, Inc., PO Box 6208, Falls Church, VA 22040.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3). The first article may be either a preproduction or initial production model.

3.2 Safety and health requirements. The torches shall comply with the general safety requirements of OSHA 2206 that are applicable to the torches themselves. Additional safety and health requirements shall be as specified (see 6.2.1).

3.2.1 UL listing. All torches furnished under this specification shall meet the requirements of ANSI/UL 123. The torches shall be approved by Underwriters Laboratories and shall be labeled to indicate their approval in accordance with UL labeling requirements.

3.3 Design. The torches shall be new and one of the manufacturer's current models. The torches shall be of the hand held oxyacetylene positive pressure type, incorporating facilities for mixing and controlling the flow of gases to the torch tip. The length of the torches shall be not less than 5 inches and not greater than 8 inches. The weight shall be not greater than 12 ounces.

3.3.1 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the torches. When only one system of measurement is acceptable, the particular system required

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shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes, and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.3.2 Reclaimed materials. The torches may contain reclaimed materials provided such materials will not jeopardize the torches intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the torches.

3.3.3 Interchangeability. All parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or torch.

3.3.4 Lubricants. Lubricants inside the torch shall not degrade or volatilize when used in a pure oxygen atmosphere at a pressure of 60 psig.

3.4 Construction. All parts used in fabrication of the torches shall be new, clean, and tree of oil and grease. In no event shall patching processes be used for reclaiming defective parts. Surfaces of all parts shall be free of burrs, defects, and dirt. Parts subject to removal or adjustment shall not be permanently deformed by the assembly process. Torches shall not be manufactured from parts reclaimed from used torches.

3.4.1 Threads. Screw threads shall comply with the requirements of FED-STD-H28. Threads for hose connections shall comply with Compressed Gas Association Standard CGA E-1, connections 020 and 021.

3.4,2 Brazing. Brazing shall be of the highest quality workmanship prevailing among commercial manufacturers normally producing torches of the types specified herein. Flux shall be cleaned from surfaces after brazing is completed. Brazing filler metal shall have a melting temperature no lower than 1000 degrees Fahrenheit (F). Filler metal shall be visibly complete around all brazed joints.

3.4.3 Machined surfaces. Gas holes and passages shall be free of burrs. Machined surface seals shall be free of nicks, burrs, and scratches.

3.5 Components.

3.5.1 Torch handle. The torch handle shall consist of a grip, gas passages, oxygen and acetylene inlet hose connections, oxygen and acetylene control valves, and an outlet connector.

3.5.1.1 Grip. The grip of the torch handle shall be knurled, grooved, or ribbed. Grips that are part of a handle not constructed with gas passages drilled or formed through a solid body handle shall be fixed to the handle with pins, screws, threads, solder, or by brazing.

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3.5.1.2 Control valves. The oxygen and acetylene control valves shall be permanently and legibly marked on the valve body, the valve control knob, or the torch body in accordance with the instructions for marking gas inlets specified in 3.5.1.3. If the valve knobs are marked, the valve stems shall not be interchangeable. Valve stems shall be equipped with packing. Each control valve seat shall provide a seal that does not allow leakage at 50 psig or more from the upstream side of the valve to the downstream side of the valve when the valve is closed with no more than 20 inch-pounds of torque applied to the knob. Valve seats shall be constructed so that seating occurs between materials of different hardness.

3.5.1.2.1 Valve bodies. The valve bodies shall be forged, cast, or constructed from bar stock. Valve bodies shall be formed separately or as integral parts of the torch handle. Valve bodies which are formed separately shall be attached to the torch handle.

3.5.1.2.2 Valve stems. Valve stems shall be fabricated of corrosion resistant metal. Valve stems shall maintain a set valve adjustment during operation of the torch and shall not be moveable by accidental brushing of the valve with a sleeve or glove. Stems of the oxygen and acetylene valves shall be assembled to prevent removal from the valve without first removing the gland nut and packing.

3.5.1.2.3 Valve packing. The valve packing shall not leak during torch operations. The packing, or alternate means, shall provide a drag equivalent to at least four inch-ounces of torque on the valve stem to aid the stem in maintaining its setting during use and to prevent the stem from being turned by accident.

3.5.1.2.4 Valve knobs. Valve knobs shall be fabricated of corrosion resistant metal. Valve knobs shall be at least 9/16 inch in diameter and shall provide a positive grip for an operator wearing welding gloves. Valve knobs shall not separate from the stem when pulled with a force of at least 70 pounds along the centerline of the stem. Valve knobs shall not loosen or separate from the stem when at least 50 inch-pounds of torque is applied to the knob in both the clockwise and counterclockwise directions.

3,5.1.3 Gas inlets. Acetylene and oxygen inlets shall have threaded connections that comply with the requirements of CGA E-l, connections 020 and 021. Each inlet shall be permanently marked to indicate acetylene or other fuel and oxygen. The minimum permissible marking shall be no less than an “A” or “F” and “O”. Hose connections shall be at the rear of the torch handle. All threads shall be clean and free of dirt. Conical seats shall be smooth, symetrical, and free of marks, nicks, and corrosion.

3.5.2 O-rings. O-rings shall not crack or show visible evidence of deterioration following exposure for 96 hours to oxygen at a pressure of 300 psig and at a temperature of 158 degrees Fahrenheit. O-rings shall provide leak free seals. The O-rings shall conform to the requirements of ASTM D 572.

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3.5.3 Welding tip assembly. The welding tip assembly shall consist of a single flame tip conforming to 3.5.3.1 and a mixing chamber which attaches to the welding torch handle with screw threads. The welding tip may be permanently attached to the mixing chamber or attached to the chamber with screw threads. All sealing and mating surfaces shall provide a leak free seal. O-rings shall not be used between the mixing chamber and the flame. The welding tip shall have a firm metal to metal joint with the mixing chamber.

3.5.3.1 Welding tips. Welding tips shall be fabricated of copper conforming to ASTM B 133 or copper-tellurium conforming to ASTM B 124 and shall have a heat conductivity of not less than 85 percent of pure electrolytic copper. Tips shall be legibly and permanently identified for size according to the manufacturer's standard commercial practice, If the welding tips are not integral with their mixing chambers, the tips shall be threaded externally to provide a positive, gas tight, metal to metal fit to the mixing chamber. Welding tips shall be of the goose neck type. Welding tips shall provide a long or bulbous flame symmetrical with the axis of the tip end. A set of three tips, consisting of one of each orifice size listed below shall be furnished with each torch.

Number drill size	Diameter (inches)
73 - 75	0.024 - 0.021
55 - 57	0.052 - 0.043
48 - 51	0.082 - 0.067

When specified (see 6.2.1), additional tips which fall within the welding range of 1/64 inch to 3/8 inch shall be furnished.

3.6 Performance.

3.6.1 Backfire and flashback. The torches with the tip and mixing chamber attached and with the oxygen and acetylene set at the manufacturer's recommended pressures shall withstand five backfires at a rate of at least one per second with no flashback.

3.6.2 Leakage. Joints, valves, O-ring seals, metal to metal mating seals, gas passages, connections, and glands shall be capable of sustaining gas pressures of at least 50 psig internally from one gas passage to another and externally from internal gas passage to external atmosphere. Valves shall sustain the gas pressures without leakage so that when the torch is submerged in clean water for five seconds, both when the valves are closed and when valves are open to permit gas passage, there will be no bubbles formed. The welding tip seats shall not allow gas leakage at maximum operating pressures, when hand tightened.

3.7 Identification marking. Each welding torch shall be permanently marked with the manufacturer's name or with a trade mark of such known character that the source of manufacture may be readily determined.

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3.8 Workmanship. Workmanship of the welding torches shall be of a quality equal to that of comparable products furnished to the commercial market.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the' contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections, tests, and examinations shall be performed in the manufacturers designated indoor test area under the ambient temperature, the relative humidity, and the air pressure existing inside the building at the time the inspections, tests, and examinations are performed.

4.4 First article inspection. When first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. Failure of the item to pass the first article examination and all tests shall be cause for rejection.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), the quality conformance inspection shall consist of the examination in 4.6, the tests in 4.7.1

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and 4.7.2, and the inspection in 4.8. Failure of the item to pass the examination, the tests, and the inspection shall be cause for rejection.

4.6 Examination. The torch shall be visually and manually examined to determine compliance with the requirements of 3.3, 3.4 thru 3.5.3.1, 3.7, and 3.8. The visual examination shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from defects. The manual examination shall include the operation of moveable parts by hand to assure proper functioning. Examination provisions may be applied at the earliest practical point in manufacture at which time it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the torches and their component parts have passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection.

4.7 Tests.

4.7.1 Leak tests. Each torch shall be tested for leaks using an inlet gas pressure of not less than 50 psig. At the manufacturer's option, any method of leak detection with a sensitivity equal to or greater than that of a water immersion bubble leak test may be used. Any leak in the torch which cannot be stopped either by tightening a valve with a torque of 20 inch-pounds or less, or by tightening a threaded connection or valve-packing nut with a wrench as well as any leak with cannot be stopped without tightening a valve packing nut to the point that the valve cannot be turned with a torque of 20 inch-pounds or less, shall be considered an incurable leak. The torches shall be tested for leaks at a pressure of not less than 50 psig as follows:

(a) With both inlet valves on the torch handle closed and no obstruction to gas passage at the front body, both inlets shall be pressurized and there shall be no valve seat leaks.

(b) With the front body plugged and both inlet valves open, both inlets shall be pressurized and there shall be no external leaks.

Any incurable leak, detected during the course of the test, sufficient to produce a bubble in ten seconds in a water immersion test shall be cause for rejection.

4.7.2 Flame tests. With a welding tip and mixing chamber attached, the torch shall be tested as follows:

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(a) The torch shall be ignited and adjusted to produce a neutral flame at the manufacturer's recommended pressures for the tip used. The flame shall be visually examined. Failure of the tip to produce a symetrical flame shall be cause for rejection. Repeated popping or backfire of the flame shall be cause for rejection.

(b) A pilot light or other equally sensitive method shall be used to check for leaks at the tip seat and mixing chamber seat. Any leak shall be cause for rejection.

(c) The end of the tip shall be tapped against a firebrick, carbon block, or other hard surface to produce a succession of five backfires at a rate of at least one per second. Any flashback, as indicated by a pale blue flame accompanied by a sustained high-pitched whistle and possible sparks issuing from the flame hole or the handle becoming hot to the touch, shall be cause for rejection (see 3.6.1). If a flashback does occur, the oxygen valve must be closed quickly because permanent damage to the torch and injury to personnel may result.

4.7.3 Valve knob pull test. The torch shall be secured in a vise or other suitable fixture. A force of at least 70 pounds, directed away from the torch along the centerline of the valve stem, shall be applied to each valve knob. Failure of any knob to withstand the 70 pound pull without loosening or coming off shall be cause for rejection (see 3.5.1.2.4).

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A or level B preservation, level A or level B packing, and marking shall be accomplished in accordance with PIIL-W-45562.

5.2 Thread protection. Gas inlet threads shall be provided with protective caps to prevent damage during shipment, storage, and handling.

6. NOTES

6.1 Intended use. The torch described in this specification is to be used for light duty welding, brazing, and heating.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.

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(b) First article, if required (see 3.1).

(c) Additional safety and health requirements, if required (see 3.2).

(d) If torch is required to be configured in a specific measurement system (US or SI), state required system (see 3.3.1).

(e) Quantity and size of additional welding tips, if required (see 3.5.3.1).

(f) First article inspection, if different (see 4.4).

(g) Quality conformance inspection, if different (see 4.5).

(h) If preservation, packing, and marking in accordance with MIL-W-45562 is required, specify level of preservation and packing (see 5.1).

6.2.2 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When a first article inspection is required, the item to be tested should be a preproduction model or the first item offered for acceptance under the contract. The contracting officer should include specific instructions in the acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Safety and health requirements. The torches shall comply with the general safety requirements promulgated under OSHA 2206 that are applicable to the torches themselves. Exceptions and additional requirements for safety and health shall be specified in detail under 6.2.1 (c).

6.5 Subject term (key word) listing.

Brazing torch
Gas torches
Oxyacetylene, cutting-welding
Torches
Torch, brazing
Torch, welding
Welding torch

6.6 Cross-reference of classification changes. Welding torch classification changes from the previous issue of this specification are as follows:

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MIL-T-18134B (NAVY)	MIL-T-18134C
Type AA Type AB	One type only

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, clue to the extensiveness of the changes.

MILITARY INTERESTS:

Custodians:
Army - AL
Air Force - 99

Review Activities:
Air Force - 84
DLA - GS

User Activities:
Navy - CG, MC, OS, YD

Preparing activity:
DLA - IP

(Project 3433-0107)

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-18134C	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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